2022 VCE Specialist Mathematics 1 (NHT) external assessment report

Specific information

This report provides sample answers or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

Question 1

$$2xe^{y-1} + x^2e^{y-1}\frac{dy}{dx} + 4e^x\frac{dy}{dx} + 4ye^x = 0$$

$$\frac{dy}{dx} = -\frac{2xe^{y-1} + 4ye^x}{x^2e^{y-1} + 4e^x}$$

$x = 1$ and $y = 2$

$$\frac{dy}{dx} = -\frac{2e + 8e}{e + 4e} = -2$$

Question 2

$z = 2\text{cis}\left(-\frac{\pi}{3}\right)$ and $w = 2\sqrt{2}\,\text{cis}\left(-\frac{3\pi}{4}\right)$

$z^3 = 8\text{cis}(-\pi)$ and $w^2 = 8\text{cis}\left(-\frac{3\pi}{2}\right)$

$$\frac{z^3}{w^2} = \frac{8\text{cis}(-\pi)}{8\text{cis}\left(-\frac{3\pi}{2}\right)}$$

$$= \text{cis}\left(-\pi + \frac{3\pi}{2}\right)$$

$$= \text{cis}\left(\frac{\pi}{2}\right) = i$$

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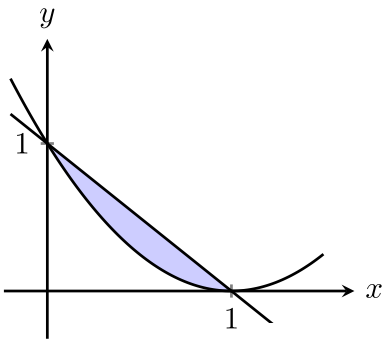
An algebraic approach using the rectangular form for complex numbers could also have been used.

$$\frac{z^3}{w^2} = \frac{\left(1-\sqrt{3}i\right)^3}{\left(-2-2i\right)^2}$$
$$= \frac{1+3\left(-\sqrt{3}i\right)+3\left(-\sqrt{3}i\right)^2+\left(-\sqrt{3}i\right)^3}{4+8i+4i^2}$$
$$= \frac{1-3\sqrt{3}i-9+3\sqrt{3}i}{8i}$$
$$= -\frac{8}{8i}$$
$$= -\frac{i}{i^2} = i$$

Question 3

Note that $y = x^2 - 2x + 1 = (x-1)^2$

The axis intercepts are at $(0,1)$ and $(1,0)$. The equation of the straight line joining the axis intercepts is $y = 1 - x$.



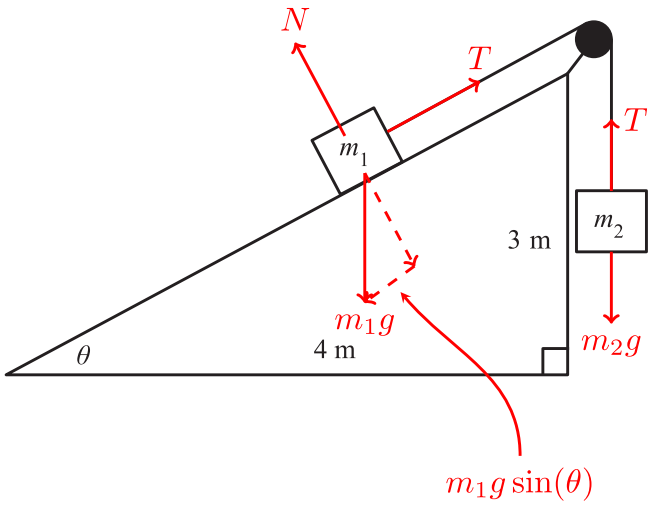
The volume is:

$$V = \pi\int_0^1 (1-x)^2\,dx - \pi\int_0^1 \left(x^2-2x+1\right)^2\,dx$$
$$= \pi\int_0^1 (x-1)^2\,dx - \pi\int_0^1 (x-1)^4\,dx$$
$$= \pi\left[\frac{1}{3}(x-1)^3\right]_0^1 - \pi\left[\frac{1}{5}(x-1)^5\right]_0^1$$
$$= \pi\left(-\frac{1}{3}(-1)\right) - \pi\left(-\frac{1}{5}(-1)\right)$$
$$= \frac{\pi}{3} - \frac{\pi}{5} = \frac{2\pi}{15}$$

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Question 4a.

Marking the forces on the diagram is recommended.



As the system is in equilibrium:

$$m_1 g \sin(\theta) - T = 0$$
$$T - m_2 g = 0$$

Therefore, $g\left(m_1 \sin(\theta) - m_2\right) = 0$.

As $\sin(\theta) = \frac{3}{5}$, it follows that:

$$\frac{3m_1}{5} - m_2 = 0 \Rightarrow m_2 = \frac{3m_1}{5}$$

Question 4b.

Using $F = ma$ gives:

$$m_1 g \sin(\theta) = m_1 a$$
$$\Rightarrow a = g \sin(\theta) = \frac{3}{5} g$$

Using the constant acceleration formula $v^2 = u^2 + 2as$:

$$6^2 = 2 \times \frac{3}{5} g \times s$$
$$\Rightarrow s = \frac{30}{g}$$

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Alternatively, an integral form for acceleration may be used:

$$v\frac{dv}{dx}=\frac{3g}{5}$$

$$\Rightarrow \int v\,dv=\int \frac{3g}{5}\,dx$$

$$\Rightarrow \frac{1}{2}v^2=\frac{3g}{5}x+c$$

When $v=0$, $x=0$. Therefore, $c=0$. When $v=6$, $x=\frac{30}{g}$.

Question 5

Let $z=x+iy$.

$$3\sqrt{x^2+(y+1)^2}=\sqrt{x^2+y^2}$$

Square both sides and complete the square to obtain:

$$8x^2+8\left(y+\frac{9}{8}\right)^2=\frac{9}{8}$$

$$\Rightarrow x^2+\left(y+\frac{9}{8}\right)^2=\frac{9}{64}$$

The centre is at $\left(0,-\frac{9}{8}\right)$ and the radius is $\frac{3}{8}$.

Question 6

$$\dot{\underset{\sim}{r}}(t)=16t^{-1/2}\underset{\sim}{i}+12t\underset{\sim}{j}-6e^{2t-8}\underset{\sim}{k}$$

$$\ddot{\underset{\sim}{r}}(t)=-8t^{-3/2}\underset{\sim}{i}+12\underset{\sim}{j}-12e^{2t-8}\underset{\sim}{k}$$

Then:

$$\ddot{\underset{\sim}{r}}(4)=-\frac{8}{8}\underset{\sim}{i}+12\underset{\sim}{j}-12\underset{\sim}{k}$$

$$=-\underset{\sim}{i}+12\underset{\sim}{j}-12\underset{\sim}{k}$$

$$\left|\ddot{\underset{\sim}{r}}(4)\right|=\sqrt{1+144+144}$$

$$=\sqrt{289}=17$$

The magnitude of the force is $2\times 17=34$.

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Question 7a.

$\frac{x}{4} = \sec(t)$, $\frac{y}{2} = \tan(t)$

As $\sec^2(t) - \tan^2(t) = 1$,

$$\left(\frac{x}{4}\right)^2 - \left(\frac{y}{2}\right)^2 = 1$$

$$\Rightarrow \frac{x^2}{16} - \frac{y^2}{4} = 1.$$

Question 7b.

$$\underset{\sim}{\dot{r}}(t) = \frac{4\sin(t)}{\cos^2(t)}\underset{\sim}{i} + 2\sec^2(t)\underset{\sim}{j}$$

$$\underset{\sim}{\dot{r}}\left(\frac{\pi}{6}\right) = \frac{4\times\frac{1}{2}}{\frac{3}{2}}\underset{\sim}{i} + 2\times\frac{4}{3}\underset{\sim}{j} = \frac{8}{3}\underset{\sim}{i} + \frac{8}{3}\underset{\sim}{j}$$

$$\left|\underset{\sim}{\dot{r}}\left(\frac{\pi}{6}\right)\right| = \sqrt{\left(\frac{8}{3}\right)^2 + \left(\frac{8}{3}\right)^2} = \frac{8\sqrt{2}}{3}$$

Question 8a.

Let S and D represent the distribution of print runs of standard and deluxe cartridges respectively.

$$\begin{aligned} \mathrm{E}(S+D) &= \mathrm{E}(S) + \mathrm{E}(D) \\ &= 2000 + 2500 = 4500 \end{aligned}$$

and

$$\begin{aligned} \mathrm{Var}(S+D) &= \mathrm{Var}(S) + \mathrm{Var}(D) \\ &= 30^2 + 40^2 = 50^2 \end{aligned}$$

Therefore, $\mathrm{sd}(S+D) = 50$.

Question 8b.

$$\mathrm{Var}(S+D+E) = 40^2 + 30^2 + (\sigma_E)^2$$

$$40^2 + 30^2 + (\sigma_E)^2 \leq 60^2$$

$$\Rightarrow (\sigma_E)^2 \leq 60^2 - 50^2 = 1100$$

Therefore, $\sigma_E \in \left(0, \sqrt{1100}\right]$.

Note that $\sigma_E > 0$, as E is a normally distributed random variable.

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Question 8c.

$z\dfrac{s}{\sqrt{n}}=\dfrac{1999.6-1960.4}{2}=\dfrac{39.2}{2}=19.6$

$z=1.96$ and $s=40$

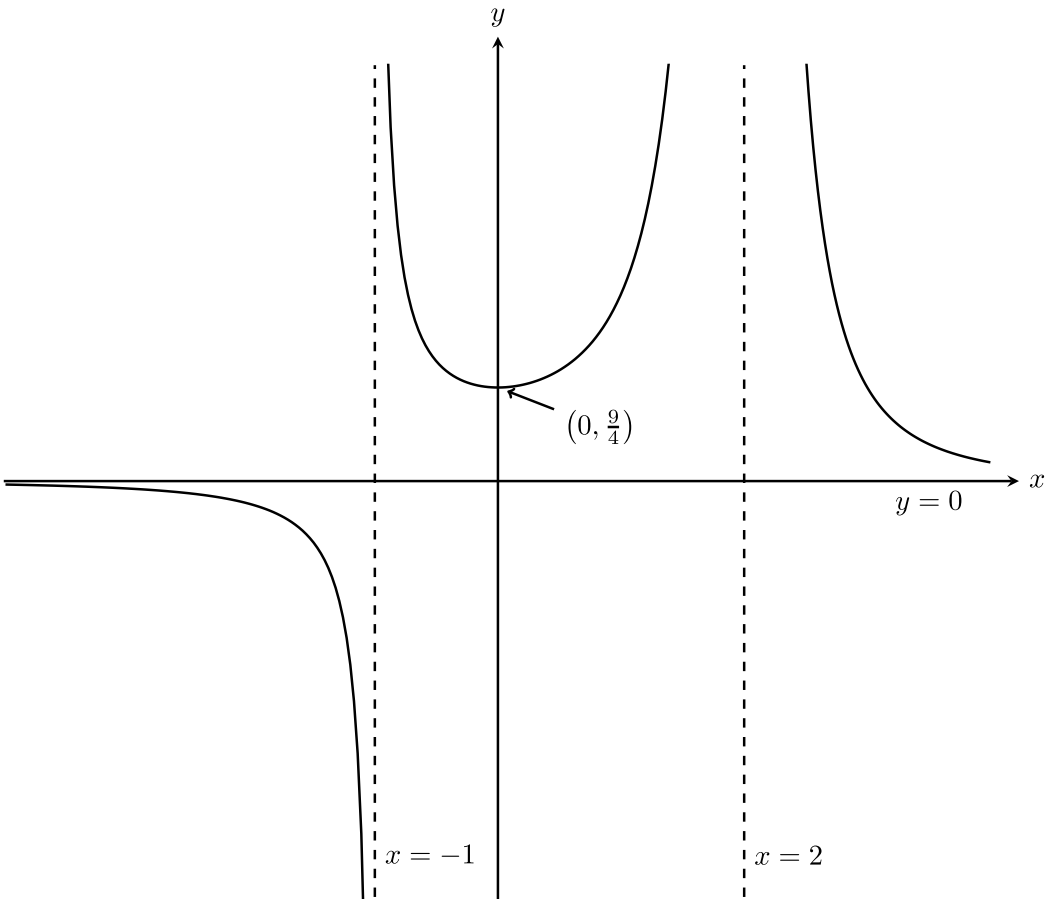
$1.96\times\dfrac{40}{\sqrt{n}}=19.6$

$\Rightarrow\dfrac{40}{\sqrt{n}}=10$

$\Rightarrow\sqrt{n}=4$

$\Rightarrow n=16$

Question 9a.



Asymptotes $x=-1$, $x=2$ and $y=0$.

Minimum turning point and y–intercept at $\left(0,\dfrac{9}{4}\right)$.

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Question 9b.

$$\frac{9}{(x+1)(x-2)^2} \equiv \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

$A=1,\ B=-1,\ C=3$

Then:

$$\int_0^1 \frac{9}{(x+1)(x-2)^2}dx = \int_0^1 \left(\frac{1}{x+1} - \frac{1}{x-2} + \frac{3}{(x-2)^2}\right)dx$$

$$= \left[\log_e|x+1| - \log_e|x-2| - \frac{3}{x-2}\right]_0^1$$

$$= \left(\log_e(2)+3\right) - \left(-\log_e(2) + \frac{3}{2}\right)$$

$$= 2\log_e(2) + \frac{3}{2} = \log_e(4) + \frac{3}{2}$$

Question 10a.

$$f'(x) = \frac{\sqrt{3}}{2}\left(3\cdot\frac{\sqrt{3}}{3}\cdot\frac{1}{\sqrt{1-\frac{x^2}{3}}} + \sqrt{3-x^2} - \frac{2x^2}{2\sqrt{3-x^2}}\right)$$

$$= \frac{\sqrt{3}}{2}\left(\frac{3}{\sqrt{3-x^2}} + \frac{3-x^2}{\sqrt{3-x^2}} - \frac{x^2}{\sqrt{3-x^2}}\right)$$

$$= \frac{\sqrt{3}}{2}\cdot\frac{6-2x^2}{\sqrt{3-x^2}}$$

$$= \sqrt{3}\cdot\frac{3-x^2}{\sqrt{3-x^2}}$$

$$= \sqrt{3}\cdot\sqrt{3-x^2} = \sqrt{9-3x^2}$$

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Question 10b.

$$\int_0^{\frac{3}{2}} \sqrt{\left(\frac{t^2}{2}\right)^2 + 9 - 3t^2}\, dt = \int_0^{\frac{3}{2}} \sqrt{\frac{1}{4}\left(t^4 - 12t^2 + 36\right)}\, dt$$

$$= \frac{1}{2}\int_0^{\frac{3}{2}} \sqrt{\left(t^2 - 6\right)^2}\, dt$$

$$= \frac{1}{2}\int_0^{\frac{3}{2}} \left(6 - t^2\right) dt$$

$$= \frac{1}{2}\left[6t - \frac{1}{3}t^3\right]_0^{\frac{3}{2}}$$

$$= \frac{1}{2}\left(9 - \frac{9}{8}\right)$$

$$= \frac{1}{2} \cdot \frac{63}{8} = \frac{63}{16}$$

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